

Rank Candidates Like a Great Recruiter

Not like a keyword filter.

0.86

Top candidate
score

21

Honeypots
detected

100%

Title-correct
top 100

<5m

100K pool
runtime

The sample submission is the wrong answer — by design.

✗ SAMPLE SUBMISSION (WRONG)

#1 HR Manager — AI keywords listed

#2 Accountant — AI keywords listed

#3 Graphic Designer — AI keywords

#4 Marketing Manager — AI keywords

#5 Content Writer — AI keywords

...

#27 AI Engineer (actual fit!) — buried

#75 Senior ML Engineer — almost missed

Pure keyword counting → keyword stuffers always win

✓ OUR v3 RANKING (CORRECT)

#1 Lead AI Eng @ Razorpay 0.8643

#2 Senior ML Eng @ Zomato 0.8563

#3 Applied Scientist 0.8492

#4 Staff ML Eng @ Paytm 0.8465

#5 Senior NLP Eng @ Ola 0.8460

...

HR Managers → score ≤ 0.001

Accountants → score ≤ 0.001

Title gate + semantic intent + availability → correct ranking

12-Phase Hybrid Scoring Engine

18%



Title & Role Gate

HARD GATE. Wrong-role candidates (HR Manager, Accountant, Mechanical Engineer) get 0.05 title score. A title gate multiplier then suppresses their entire score — not just the 18% slice — to prevent career/behavioral signals from rescuing them.

22%



Skill Depth (Fuzzy)

P2: Fuzzy substring matching — 'sentence-transformers' matches 'sentence transformer'. Depth = proficiency × endorsements × duration_months, not just presence. Keyword stuffers (20 skills, avg 2 endorsements, 3-month duration) get 65% penalty.

30%



Career + Semantic Intent

P3/P10: Searches headline, summary, titles, descriptions for search/ranking/matching CONCEPTS. Finds people who built ranking systems but wrote 'marketplace personalization' not 'FAISS'. Gated by technical title to prevent SEO writers from matching. P4: company normalization (Google India → google).

20%



Behavioral Signals

P5: Softened multiplier — $\text{base} \times (0.75 + 0.25 \times \text{availability})$. Ghost candidates are down-weighted, not destroyed. 23 Redrob signals: recency, response rate, open-to-work, GitHub score, interview completion, notice period, skill assessments.

3%



Education

Pre-tiered (tier_1–tier_4). Low weight — JD doesn't emphasize education. Field and degree bonuses.

2%



Location (4-tier)

P9: Preferred cities (Pune/Noida/Bangalore) → 1.0. Willing to relocate → 0.90. Other Indian city → 0.75. Outside India → 0.60.

Every change measurably improves ranking quality

P2

Fuzzy Skill Matching

Before: `skills = {s['name'].lower()}; if s in RETRIEVAL_SKILLS`

After: `fuzzy_match(skill_name, keywords)` — substring, not exact set

→ Catches 'sentence-transformers', 'dense retrieval', 'OpenAI embeddings'

P3+P10

Semantic Intent Score

Before: Only checked skills list for tool names (FAISS, Pinecone)

After: Searches all text for concepts: ranking, marketplace, personalization, matching

→ Surfaces engineers who built search systems without naming the tools

P4

Company Normalization

Before: 'Google India' didn't match tier-1 company 'google'

After: `normalize_company()` → Google India / Google LLC / Google Cloud → 'google'

→ Tier-1 signals now work for all company name variants

P5

Softened Availability Multiplier

Before: $\text{final} = \text{base} \times \text{availability}$ → ghost candidate score crushed to 0.08

After: $\text{final} = \text{base} \times (0.75 + 0.25 \times \text{availability})$ → minimum multiplier 0.77

→ Down-weight unavailable candidates, don't destroy them

P7+P8

Softer Penalties

Before: CV/speech/robotics → -70% skill, -50% career. Consulting only → 0.10

After: Only penalize wrong-domain if NO retrieval/NLP/search signal anywhere

→ P8: consulting-only → 0.45. Cross-domain engineers no longer unfairly penalized

P9

4-Tier Location

Before: Preferred city → 1.0, else → 0.35 (binary)

After: Preferred → 1.0, relocate → 0.90, other India → 0.75, abroad → 0.60

→ Relocatable candidates no longer unfairly equal to out-of-country candidates

Down-weight. Don't destroy.

v2: Final = Base × Availability

v3: Final = Base × (0.75 + 0.25 × Availability)

Profile	Avail Raw	v2 ×	v3 ×	v2 Final	v3 Final	Winner
★ Lead AI Eng, Active	0.94	0.94	0.98	0.865	0.906	v3 ✓
✓ Good Fit, Slow Resp	0.65	0.65	0.91	0.488	0.684	v3 ✓
◆ Great Skills, Ghost	0.18	0.18	0.80	0.148	0.652	v3 ✓
💀 Perfect Paper, Dead	0.08	0.08	0.77	0.070	0.678	v3 ✓

The ghost candidate (0.08 avail) now gets ×0.77 instead of ×0.08 — a fairer signal.

Top 15 from the real 100,000-candidate pool

100K
candidates scored

21
honeypots removed

0.8643
top score

1.000
avg title fit top 100

<5min
runtime (CPU)

#	Candidate	Role & Company	Score	Reasoning (excerpt)
1	CAND_0081846	Lead AI Engineer @ Razorpay	0.8643	RAG-based ranking pipeline, 50M+ queries/mo
2	CAND_0018499	Senior ML Engineer @ Zomato	0.8563	RAG ranking pipeline, recruiter-facing search product
3	CAND_0071974	Applied Scientist	0.8492	Production recommendation system, offline→A/B test @ Mad Street Den
4	CAND_0077337	Staff ML Engineer @ Paytm	0.8465	Production recommendation, A/B test @ Paytm marketplace
5	CAND_0008425	Senior NLP Engineer @ Ola	0.8460	Large-scale semantic search, 35M+ items corpus
6	CAND_0002025	Senior AI Engineer @ Apple	0.8346	Production recommendation, marketplace A/B test
7	CAND_0055905	Senior ML Engineer @ Flipkart	0.8342	Semantic search 35M+ items, Flipkart tier-1
8	CAND_0079387	ML Engineer @ Microsoft	0.8342	Content recommendation 10M+ users, collaborative filtering + ranking
9	CAND_0041610	Recom Systems Eng @ Zoho	0.8071	E-commerce search, hand-tuned → learning-to-rank
10	CAND_0052328	ML Engineer @ Amazon	0.8061	RAG pipeline: chunking, OpenAI embeddings, Pinecone, answer generation

21 honeypots correctly neutralized from 100K pool



Wrong Title + AI Keywords

🔍 Signal: HR Manager / Accountant with full AI skill list

⚙️ How: Title gate: $\text{score} \times (0.15 + 0.85 \times \text{title_score})$. $\text{title_score}=0.05 \rightarrow \text{gate}=0.19$

✓ Result: Final score ≤ 0.019 regardless of skills or career signals



Keyword Stuffer

🔍 Signal: >12 skills, avg endorsements < 4 , avg duration < 8 months

⚙️ How: Skill score $\times 0.35$ penalty. 20 AI skills at 0 endorsements = raw $\times 0.35$

✓ Result: Skill component collapses from 0.80 $\rightarrow$ 0.28



Impossible Profile (Honeypot)

🔍 Signal: 5+ expert-level skills with 0 duration_months or 0 endorsements

⚙️ How: `check_honeypot()` $\rightarrow$ score = 0.0001 (hard floor)

✓ Result: 21 such profiles detected and floored in the 100K pool



Behavioral Twin / Ghost

🔍 Signal: Strong ML profile, 200+ days inactive, 5% response rate

⚙️ How: P5 multiplier: $0.75 + 0.25 \times 0.08 = 0.77$ (v3) vs 0.08 (v2)

✓ Result: v3: down-weighted not destroyed — still ranks below active peers

Specific facts from actual profiles, not templates

✗ v2 REASONING (GENERIC)

"Senior ML Engineer with 8.1 yrs; strong in elasticsearch, information retrieval; response rate 86%, active 12d ago, notice 30d."

Problems: Generic template. No career specifics. Could apply to any ML engineer.

✓ v3 REASONING (SPECIFIC)

"Built a RAG-based ranking pipeline serving 50M+ queries per month for an internal recruiter-facing search product at Razorpay (6.7 yrs as Lead AI Engineer). open to work; 94% recruiter response rate; active 8d ago; 30d notice; GitHub score 82; based in Bangalore."

Why better: Real career achievement. Quantified scale. All signals in one sentence.

How v3 builds reasoning:

1. Scan all career descriptions for sentences containing retrieval/ranking/production keywords
2. Pick the sentence with the highest signal-word hit count
3. Append: company name, title, YoE as context
4. Sentence 2: open-to-work flag, recruiter response %, days active, notice period, GitHub score, location

HOW TO RUN

3 commands. No GPU. No network. No pip install.

terminal

```
# 1. No install required - pure Python stdlib
```

```
# 2. Run on the full 100K candidate pool
```

```
python rank.py \  
  --candidates candidates.jsonl.gz \  
  --out my_team.csv
```

```
# Output:
```

```
✅ Scored 100000 candidates (21 honeypots)
```

```
📊 Score range: 0.8643 → 0.7126
```

```
🏆 Top: Lead AI Engineer @ Razorpay
```

```
# 3. Validate format before submitting
```

```
python validate_submission.py my_team.csv
```

```
Submission is valid. ✓
```

Key Properties

Pure Python stdlib

No pip install. Runs anywhere Python 3.8+ is installed.

~4 min on CPU

100K candidates, single-threaded. Well within the 5-min constraint.

Format guaranteed

Score non-increasing, tie-breaks by candidate_id. Passes validator every time.

Trap-aware

4 distinct honeypot patterns detected. 21 flagged in real pool.

Explainable

Every row has specific 2-sentence reasoning from real profile data.

Fully offline

No API calls, no network during ranking. Reproducible on any machine.

THE OUTCOME

A shortlist a recruiter can trust.

0.8643

Top candidate
score (v3)

21

Honeypots
neutralized

100%

Correct titles
in top 100

✓

Submission
validated